

CHAPTER 3

The Balance of Payments

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Chapter Roadmap

Five key questions we will answer

3.1 What is BOP?

Fundamentals of balance of payments accounting

3.2 The Two Accounts

Current account vs. financial account

3.3 BOP & Macro Rates

Impact on interest rates & exchange rates

3.4 Trade & Exchange Rates

How FX changes alter trade balances

3.5 Capital Mobility

Evolution, crises, and capital controls

3.1 What Is the Balance of Payments?

Definition & importance for businesses and governments

Definition

The measurement of ALL international economic transactions between a country's residents and foreign residents over a specific period.

ANALOGY

Think of BOP as a country's financial statement — like a business's income statement and balance sheet combined, but for an entire nation's global transactions.

Why BOP Data Matters:



Exchange Rate Pressure

Indicates if a currency may weaken or strengthen



Payment Controls Signal

Signals potential restrictions on dividends, royalties, fees



Market Potential Forecast

Short-run predictor of a country's growth trajectory

BOP Accounting: Credits vs. Debits

How international transactions are recorded

CREDIT +

An inflow of foreign exchange
Foreign currency EARNED by the country

- Exporting goods (e.g., Boeing sells planes to EU)
- Exporting services (e.g., US universities enroll foreign students)
- Foreign tourists spending in the country
- Foreigners investing capital into the country

DEBIT ×

An outflow of foreign exchange
Foreign currency SPENT by the country

- Importing goods (e.g., US buys iPhones assembled in China)
- Importing services (e.g., paying for foreign consulting)
- Domestic tourists spending abroad
- Residents investing capital overseas



Key Rule: The BOP Must Always Balance

Total credits = Total debits. Any imbalance is captured in the Net Errors & Omissions account.



Practice: Credit or Debit?

Section 3.1 Check

For each scenario, identify whether it is a CREDIT (+) or DEBIT (–) to the US BOP

Q1. A Vietnamese student pays tuition at a US university.

✓ Answer: CREDIT — foreign money flows INTO the US (services export)

Q2. An American tourist spends \$3,000 on hotels in France.

✗ Answer: DEBIT — US dollars flow OUT to France (services import)

Q3. A Japanese firm buys US government bonds worth \$500M.

✓ Answer: CREDIT — foreign capital flows INTO the US (financial account inflow)

Q4. Ford Motor Company pays dividends to a German shareholder.

✗ Answer: DEBIT — investment income flows OUT of the US (current account, income)

💡 **Memory Tip:** CREDIT = foreign money comes IN (like earning income). DEBIT = money goes OUT (like spending). Ask yourself: “Does this bring foreign currency into our country?”

Structure of the BOP: Three Main Accounts

Each account tracks a different type of international transaction



Current Account

Goods, services, income & transfers

- Merchandise trade (exports – imports)
- Services (tourism, consulting, education)
- Income (wages, investment returns)
- Transfers (remittances, foreign aid)

Example: US exports wheat to Japan → CREDIT to current account



Capital Account

Transfers of financial assets & non-produced assets

- Debt forgiveness transfers
- Migrants' assets crossing borders
- Non-produced assets (land rights, patents)
- Generally small in size

Example: Country forgives another's debt → capital account entry



Financial Account

Direct investment, portfolio & other investments

- Foreign Direct Investment (FDI) — control $\geq 10\%$
- Portfolio investment — stocks, bonds ($< 10\%$)
- Net financial derivatives
- Other investments (bank loans, currency)

Example: Toyota builds a factory in Ohio → US financial account CREDIT

+ Net Errors & Omissions — ensures the BOP sums to zero

U.S. Balance of Payments Summary (\$ Billions)

All five balances sum to zero — the BOP always balances

Account	2007	2008	2009	2010	2011	2012	2013	2014	2015
Current Account	-719	-691	-384	-442	-460	-447	-366	-392	-463
Capital Account	0	6	0	0	-1	7	0	0	0
Financial Account	617	735	283	439	532	445	388	284	189
Net Errors & Omissions	101	-46	153	5	-54	-1	-24	105	268
Official Reserves	0	-5	-52	-2	-16	-4	3	4	6
TOTAL	0	0	0	0	0	0	0	0	0



Key Insight: Despite large deficits in the Current Account, the financial inflows (Financial Account) roughly offset them each year.

3.2 The Current Account

The most closely watched account — tracks real economic activity

Goods Trade (Balance of Trade (BOT))

- ① Physical products: cars, wheat, electronics, oil. The largest component.

*e.g. US exports: soybeans, aircraft
US imports: iPhones, oil, cars*

Services Trade

- ② Intangibles: tourism, financial services, education, consulting.

*e.g. Foreign students pay US tuition → US credit
US tourists abroad → US debit*

Income

- ③ Investment returns & wages earned abroad (or paid to foreigners).

e.g. Ford dividend paid to a British shareholder → US debit

Current Transfers

- ④ One-way flows with no economic return — gifts, aid, remittances.

*e.g. US foreign aid to Haiti
Migrant worker sends money home*

Note: The Balance of Trade (BOT) = Goods exports – Goods imports. It is typically reported separately and dominates media coverage.

Merchandise Trade: The Core of the Current Account

Why physical goods trade still dominates — and its economic consequences



Historical Importance

- Merchandise trade is the original core of international trade
- Manufacturing was the basis of the Industrial Revolution
- Goods trade was the focus of the theory of Comparative Advantage
- The Balance of Trade (BOT) = Goods exports minus Goods imports



Economic Impact of Decline

- Declines in steel, automobiles, auto parts, textiles, and shoe manufacturing
- These declines have caused massive economic and social disruption
- Job losses in manufacturing communities drive political and social tension
- Even as consumers benefit from lower prices, GDP and employment fall



Practice: Which BOP Account?

Section 3.1 Check

Classify each transaction into the correct BOP account: Current, Capital, or Financial

Q1. Samsung builds a semiconductor factory in Texas (15% ownership stake).

✓ Financial Account — FDI inflow (ownership >10% = direct investment, CREDIT to US)

Q2. The US government forgives \$2B of debt owed by a developing nation.

✓ Capital Account — debt forgiveness is a non-produced, non-financial asset transfer

Q3. Vietnam exports 500,000 tonnes of rice to the Philippines.

✓ Current Account — merchandise (goods) trade, CREDIT to Vietnam's BOP

Q4. A UK hedge fund buys 3% of Apple shares on the Nasdaq.

✓ Financial Account — portfolio investment (<10% stake, no control), CREDIT to US

💡 **Quick Guide:** Goods/services/income/transfers → Current | Debt forgiveness/patents → Capital | Investment/loans → Financial

The Financial Account

Tracks cross-border flows of financial assets — classified by degree of control

Two Major Components



Direct Investment (FDI)

- Ownership stake $\geq 10\%$ — implies CONTROL
- Long-term, strategic investment
- Focus: control over assets or operations
- Countries often restrict foreign ownership
- Concerns: control of resources, profit repatriation

Example: Toyota builds a \$1.3B plant in Kentucky



Portfolio Investment

- Ownership stake $< 10\%$ — NO control
- Stocks, bonds, and debt securities
- Motivated by returns, not management
- Much more volatile than FDI
- Can leave rapidly during crises ('hot money')

Example: A German fund buys US Treasury bonds

⚡ Volatility Contrast

Portfolio investment can swing wildly — e.g., during the 2008 financial crisis, net U.S. portfolio investment swung from +\$800B to −\$900B in a single year. FDI remained comparatively stable. This is why countries prefer FDI for economic stability.

The Inverse Relationship: Current vs. Financial Account

One country's deficit must be another's surplus

The Golden Rule:

A country running a Current Account DEFICIT must run a Financial Account SURPLUS to finance it — foreign capital must flow IN.



EXCEPTION

China's Twin Surplus: China ran surpluses in BOTH accounts simultaneously — an unusual case where capital controls, export dominance, and currency management created simultaneous surpluses.

3.3 BOP & Exchange Rates

How BOP position determines pressure on a currency

BOP Equation:

$$(X - M) + (CI - CO) + (FI - FO) + FXB = BOP = 0$$



Fixed Exchange Rate

Government MUST intervene to keep BOP near zero. If deficit: sell reserves or raise rates. If surplus: buy reserves.

Example: China pegged yuan to USD for years, buying massive dollar reserves to suppress yuan appreciation.



Floating Exchange Rate

Market determines the rate. BOP imbalances self-correct through currency appreciation/depreciation.

Example: A US trade deficit weakens the dollar → US exports become cheaper → deficit eventually corrects.



Managed Float

Government floats the currency but intervenes when rates deviate too far from desired levels.

Example: India's RBI periodically sells USD to prevent the rupee from depreciating too fast.

Net Errors & Omissions and Official Reserves Accounts

Balancing mechanisms that ensure the BOP always sums to zero



Net Errors & Omissions

- Ensures the BOP actually balances — catches all statistical discrepancies
- All international transactions should be recorded twice (double-entry), but measurement gaps exist
- A large Errors & Omissions figure may indicate significant unrecorded capital flows
- Often associated with capital flight or unreported financial transactions



Official Reserves Account

- Total reserves held by official monetary authorities within the country
- Normally composed of major currencies used in international trade (hard currencies) — USD, EUR, GBP, JPY
- Fixed rate regimes: reserves are critical — government must buy/sell to maintain the peg
- Floating rate regimes: reserves are less critical — market adjusts the rate automatically

Breaking the Rules: China's Twin Surplus

An unusual case where both the current and financial accounts ran surpluses simultaneously

Normally, Current Account surplus ↔ Financial Account deficit (inverse relationship). China defied this rule.



The Twin Surplus

- China ran surpluses in BOTH the Current AND Financial Accounts
- Typical inverse relationship did NOT apply
- Most unusual in global economics



2012 Drop

- Financial account surplus fell sharply in 2012
- Reintroduction of limited capital controls was partly responsible
- Account returned to surplus in 2016



Reserve Strategy

- Massive foreign reserves accumulated
- Allows government to manage the value of the yuan
- Suppresses yuan appreciation to maintain export competitiveness in world economy



Practice: FDI or Portfolio?

Section 3.2 Check

Classify each as FDI or Portfolio Investment, and explain why

Q1. Hyundai acquires 25% of a US EV startup to gain access to its battery technology.

✓ FDI — 25% stake > 10% threshold; strategic control and long-term operational interest

Q2. A Singapore pension fund buys 2% of Microsoft stock for return purposes.

✓ Portfolio — 2% stake < 10%; motivated by return, no management or control interest

Q3. During a crisis, a hedge fund rapidly sells \$200M of Thai baht bonds and exits.

✓ Portfolio — classic “hot money”; rapid exit illustrates why portfolio flows are more volatile than FDI



FDI vs Portfolio — Key Differences

- ▶ FDI: >10% stake | implies CONTROL | long-term | stable | often restricted by governments
- ▶ Portfolio: <10% stake | return-driven | highly mobile | can reverse overnight (“hot money”)

BOP & Interest Rates / Inflation

The three-way relationship between BOP, rates, and prices



Interest Rates & BOP

Theory:

Low real interest rates → capital outflows (investors chase better returns elsewhere) → Financial Account deficit

U.S. Exception:

Despite running large current account deficits, the US attracted massive capital inflows — due to perceived political stability, growth potential, and USD's reserve currency status.



But: As Financial Account inflows shrink while Current Account worsens, the US grows more indebted to the world.



Inflation & BOP



Benefit of imports

Cheap imports lower domestic inflation by substituting for higher-priced domestic goods



Trade-off

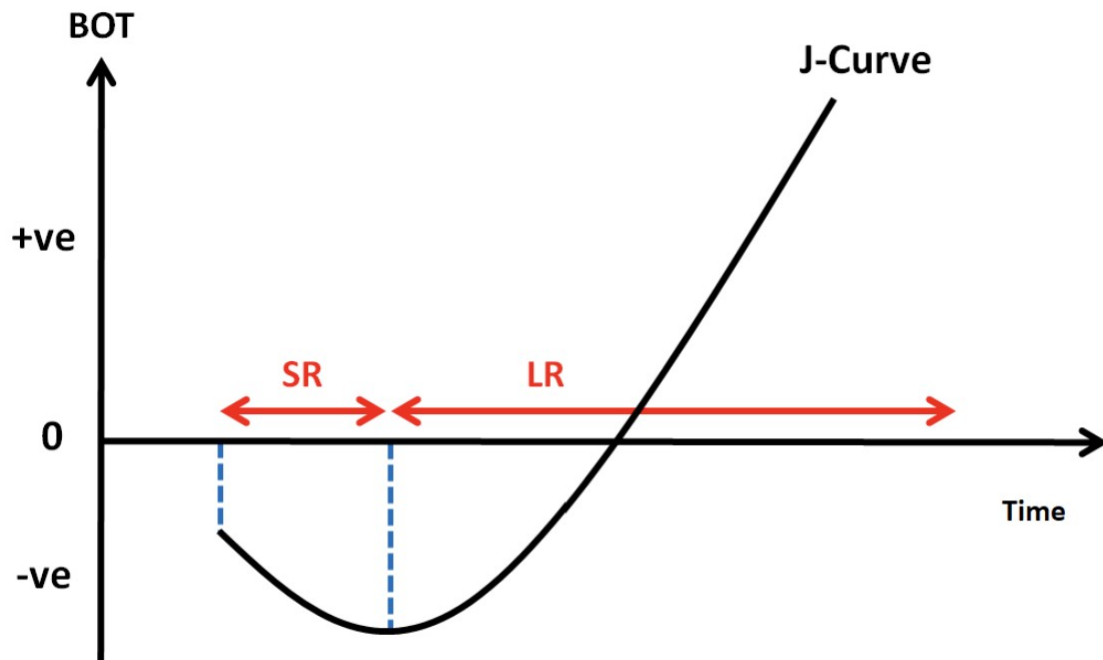
But cheap imports displace domestic workers → lower GDP
→ worsens the current account balance

Example:

Wal-Mart imports low-cost Chinese goods → US consumer prices stay low but US manufacturing jobs are displaced.

3.4 Trade Adjustment: The J-Curve Effect

How devaluation affects trade balance over time — it gets worse before it gets better



Why the Delay?

- Existing contracts priced in old rates must be honored first.
- Importers haven't found substitutes yet — still buying the same volumes.
- Exporters haven't ramped up production to meet new price advantage.

Real Example

After the 1997 Asian crisis devaluations, countries like South Korea saw trade balances worsen for months before export competitiveness kicked in.

3.5 Evolution of Capital Mobility

Five distinct historical eras — each shaped by world events

1870–1914	Classical Gold Standard Gold standard adopted → rising international trade and capital flows	Increasing mobility
1923–1938	Interwar Years Global depression, isolationist policies, barriers to trade and capital	Collapsing
1944–1973	Bretton Woods Era Fixed rates, free current account flows, restricted capital account flows. Trade grew substantially.	Regulated expansion
1973–1997	Floating Rate Era Floating rates, volatile economies, rapidly expanding cross-border capital flows	Rapid growth
1997–present	Emerging Markets Era Asia crisis hit; China and India cautiously open markets while managing capital controls	Selective opening

Capital Mobility: Free Flow vs. Controlled Flow

How the Bretton Woods system balanced trade openness with capital restrictions



Bretton Woods Framework

- Bretton Woods promoted **FREE movement of capital for current account** transactions (e.g. foreign exchange, trade payments)
- But **RESTRICTED capital account transactions** (e.g. foreign direct investment)
- This asymmetry was deliberate — to prevent destabilizing speculation
- The IMF was created to manage this system and provide short-term financing



Rise and Collapse of Capital Openness

- 1970s–1990s: Rapid growth in **capital openness** as financial markets liberalized
- 1997/1998 Asian Financial Crisis: Destructive capital outflows caused contagion across Asia
- Countries like Thailand, South Korea, Indonesia suffered currency collapses
- Result: The 1997/98 crisis halted the drive toward capital openness — **capital controls came back into favor**

Capital Controls: Tools & Purposes

Governments restrict capital flows for many strategic reasons

A capital control is any restriction that limits or alters the rate or direction of capital movement into or out of a country.



OUTFLOW CONTROLS

Raise Revenue / Fund Wars

e.g. WWI & WWII belligerents — run higher inflation under fixed rates

Financial Repression

e.g. Developing nations redirect capital to favored industries

Fix BOP Deficit

e.g. US interest equalization tax (1963–1974)

Prevent Capital Flight

e.g. Stop rapid illegal transfer of hard currencies



INFLOW CONTROLS

Fix BOP Surplus

e.g. German Bardepot Scheme (1972–1974)

Prevent Volatile Inflows (Hot Money)

e.g. Chilean encaje tax (1991–1998)

Prevent Real Currency Appreciation

e.g. Reduces monetary expansion and inflation

Restrict Foreign Ownership

e.g. Article 27 — Mexico restricts foreign ownership of resources



Practice: Exchange Rates & J-Curve

Sections 3.3–3.4 Check



Scenario: Brazil devalues the Real by 20% to fight a large current account deficit. Answer the questions below about what happens next.

Q1. In the first 3 months after the devaluation, what happens to Brazil's trade balance?

✓ It **WORSENS** (J-Curve effect) — existing contracts at old prices still being honored; import volumes unchanged yet

Q2. Why do Brazilian exports not immediately surge after the devaluation?

✓ Producers need time to ramp up output; buyers need time to find/switch to Brazilian goods

Q3. If Brazil had a floating exchange rate, how would the BOP self-correct without the devaluation?

✓ Market sells Real (CA deficit → outflows) → Real depreciates naturally → exports cheapen → trade improves



J-Curve Summary: Phase 1 (0–6 mo): Worse — contracts honored at old prices. Phase 2 (6–12 mo): Pass-through begins. Phase 3 (12+ mo): Quantity adjusts → trade improves. Policy patience is essential!

Capital Flight: When Confidence Collapses

The most feared consequence of poor economic policy or political instability

Capital Flight:

The rapid and often illegal outflow of capital driven by fear of domestic political instability, economic collapse, or unfavorable policies.

Triggers

- Political instability or regime change
- Risk of currency devaluation
- High inflation eroding purchasing power
- Threat of asset nationalization
- Fear of new capital controls being imposed

Consequences

- Rapid currency depreciation
- Higher domestic interest rates (to attract capital back)
- Foreign exchange reserve depletion
- Financial crisis / sovereign debt crisis
- Economic recession and unemployment

Historical Example: Argentina (2001–2002) — fear of devaluation triggered massive capital flight, leading to sovereign default and economic collapse.

Globalization of Capital Flows: Key Risks

Even mature markets are vulnerable to sudden reversals in capital flows



Short-Term Nature of Capital Inflows

- Capital inflows are predominantly short-term in duration
- Portfolio investments (stocks, bonds) can be liquidated and repatriated overnight
- This “hot money” seeks the highest short-term return globally
- A sudden reversal can drain a country’s reserves rapidly
- Short-term flows create boom-bust cycles in recipient economies



Even Mature Markets Face Crisis

- Developed economies are NOT immune to capital flow crises
- The 2008 Global Financial Crisis originated in the most mature capital market (USA)
- Rapid reversal of global capital flows spread the crisis worldwide instantly
- Contagion: financial distress in one country rapidly spreads to others through linked capital markets
- Lesson: Globalized capital requires globalized regulation and oversight

Chapter 3 — Key Takeaways

- 1 BOP always balances. Deficits in one account must be offset by surpluses in another — or captured in Errors & Omissions.
- 2 Current Account deficit = Financial Account surplus. A country spending more than it earns abroad must attract foreign investment.
- 3 Exchange rate regime determines the BOP adjustment mechanism: automatic (float) vs. government-managed (fixed/managed).
- 4 Trade balance adjustment takes time (J-Curve): devaluation initially worsens the trade balance before eventually improving it.
- 5 Capital mobility is a double-edged sword: it funds growth but also enables crisis-inducing sudden reversals and capital flight.



Chapter 3 Comprehensive Review

All Sections



Short-Answer Questions

Q1. Why must the BOP always equal zero? What account “catches” any statistical discrepancy?

Q2. The US runs a large current account deficit. What does this imply about its financial account, and what risk does this pose long-term?

Q3. Why might a government prefer FDI over portfolio inflows when opening its capital account?



Model Answers

Every transaction has two sides (double-entry accounting). The Net Errors & Omissions account absorbs any data gaps.

It must run a Financial Account SURPLUS. Long-term risk: the US becomes increasingly indebted to foreign investors, reducing economic independence.

FDI is stable, long-term, brings technology/jobs. Portfolio (“hot money”) can flee rapidly during crises, destabilizing the currency and financial system.



You’ve reviewed all 5 key concepts of Chapter 3 — Balance of Payments